Cyber-Physical Dam Integration for Minimization of Emergency Overflow

By: Jasper Agrawal, Will Dignam & Aidan Visconti

Department of Civil and Environmental Engineering

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John D. Albertson

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Abstract:

Modern American dams are in poor condition, receiving a "D" rating in the 2021 ASCE infrastructure report, which grades infrastructure on a standard letter scale. Over the past two decades, the number of dams classified as high-hazard has doubled, and overuse of emergency spillways remains the main cause of dam failure nationwide. These failures cause large downstream flooding, ecosystem damage, and energy loss, affecting communities and infrastructure that depend on dams for flood control and power. To address this, the present study proposes a cyber-physical system (CPS) framework for dam management that utilizes sensors, hydrologic modeling, and forecast-based gate control to optimize reservoir and turbine operations and reduce reliance on emergency spillways. Using HEC-HMS and hydrographs to model the watershed upstream of Moore Dam in Littleton, NH, baseline simulations of 10, 50, and 100-year storm events revealed that static gate conditions (without optimization) produce peak spillway discharges of approximately 46,500, 101,000, and 137,000 cfs, respectively, with pool elevations climbing well above normal operating elevations in all three cases. A forecast-based LP program was then developed that demonstrates that CPS-enabled gate control can quantitatively reduce peak outflow and minimize emergency spillway usage compared to no system optimization. These results indicate that using cyber-physical systems and predictive hydrologic modeling in dam operations is a viable strategy for improving dam safety and flood risk management across the United States.

Introduction & Literature Review:

For many years, dams have played a vital role in flood prevention, energy production, and local water supply. However, these dams must be effectively managed and maintain their structural stability during severe hydrological events. This point has been stressed by recent studies, which have revealed that numerous dams in the United States are becoming old, the number of high-risk facilities are increasing, and that the threat of failures in spillways is a major concern for communities living downstream ("Cybersecurity in the Age of Digital Dams"). These issues have inspired the introduction of innovative management approaches, with cyber-physical systems receiving significant attention in the context of water infrastructures. In literature surrounding water-based infrastructure, solutions are expected to integrate physical assets with advanced sensors, communication technologies, and decision-support capabilities to provide managers with the necessary flexibility for responding to variable conditions based on more dynamic operating policies ("Cybersecurity in the Age of Digital Dams"). It is particularly important for research findings to be applicable to dams due to the massive impact even a slight change in reservoir level could have on spillway flows (Khadka). In the case of the present study, Moore Dam, located in Littleton, New Hampshire, will be studied, due to the fact that it acts both as a hydropower station and reservoir routing facility, meaning that storm activities would influence the flood characteristics as well as power generation (National Inventory of Dams).

Decision-making based on physical processes is a major pillar of hydraulic and hydrological engineering. Hydraulics in open channels and control structures demonstrate that the discharge characteristics of outlets depend heavily on water level, channel dimensions, and flow conditions. As a result, spillway operation does not represent a simple on/off process but rather a non-linear process response (Khadka). Rainfall-runoff modeling and routing techniques in reservoirs play a key role in assessing how storm precipitation translates into inflow hydrographs, making them particularly useful in forecast-based management (US Geological Survey).

The significance of accurate forecasts is further amplified by some of the notable cases of dam failures. For instance, in the paper authored by Bunn et al. (2022), the meteorological conditions leading up to the failure of the Oroville Dam in 2017 were such that more than 600 mm of rainfall had occurred in the watershed of the Feather River located upstream from the dam in the period from February 2-11, 2017. In their analysis, the authors point out that the primary spillway, which was already operational due to high water levels, suffered damage on February 7, and the emergency spillway was used for the first time in the dam's history on February 9(Bunn et al., 2022). The importance of this case lies in the fact that it is not only related to structural issues, but also to forecasting. As discussed in the paper, the main challenge in this case was created by dam operators, whose forecasting was not accurate and eventually resulted in underestimating the inflow to Lake Oroville, highlighting the connection between storm data and reservoir management (Bunn et al., 2022). The study of the Oroville Dam contributes to the primary rationale of this report and shows that improved forecasting allows for preparing reservoirs before an emergency occurs. Through their use of High-Resolution Rapid Refresh model data input to the Gridded Meteorological Ensemble Tool, the authors illustrate how dynamic forecasting data could have provided better precipitation estimates during the Oroville incident, particularly when precipitation amounts were high and accuracy was most critical for effective flooding control (Bunn et al., 2022). The relevance of this to the present study is that forecast-based gate control and accurate inflow models based on precipitation data are equally critical in the improvement of this hydraulic infrastructure.

Other relevant research and literature on flood management has resulted in very similar conclusions. The Sigma Plan of Belgium is cited as one of the primary examples of comprehensive flood protection due to its integration of conventional engineering techniques with the spatial planning approach, ecological rehabilitation, and adaptive measures that enhance the effectiveness of the system over time (Climate-ADAPT, 2025). This change of perspective on flood management is characteristic of contemporary practices in water resources management, as opposed to previous approaches where it was believed that fixed infrastructure would help reduce risks of flooding altogether (Climate-ADAPT, 2025).

Additionally, according to the paper authored by Giupponi et al. (2024), the Venetian flood defense system should not just be considered a barrier system, but rather as an infrastructure network, which is successful because of the appropriately managed parameters such as time of operation, cost efficiency, and ability to adjust to future uncertainties. Therefore, this literature shows that any major flood defense facilities can be efficient only in cases of the proper decision-making concerning their use (Giupponi et al., 2024). While Moore Dam bears little resemblance to other regional systems such as the Sigma Plan in Belgium or mobile barriers in Venice, the authors and researchers involved in these studies provide valuable theoretical findings. Both cases indicate that predictive data, effective management, and adaptability are crucial for resilient infrastructure in the modern world, regardless of how much the physical systems differ from each other (Giupponi et al., 2024). In the context of this report, forecast-oriented dam operation can be considered an example of many contemporary trends in engineering that involve preemptive control rather than resistance.

The case studies presented within the relevant literature create an adequate rationale behind the current research. Assessments at the national level demonstrate that there is a need to improve dam performance through enhancing their resilience, and hydraulic references help in building up the model to describe a storm's reaction. The collapse of the Oroville dam shows the practical results and consequences of forecast uncertainty in case of high inflow rates (American

Society of Civil Engineers, 2021; Khadka; Bunn et al., 2022). Moore Dam becomes a unique example because, unlike in some other cases, the facility does not merely need to deal with floods and manage them effectively; rather, it must treat the storm water as both an operational threat and also a resource that needs to be utilized effectively in the process of hydroelectric power production. Therefore Moore Dam must be designed to carefully control its operations and releases, balancing storm flows and turbine use (Great River Hydro).

Methods:

A cyber-physical dam management system requires a very precise understanding of the physical system it is meant to run. For Moore's Creek Dam, over 25 individual parameters shape the system and watershed behavior, grouped into five functional categories: dam geometry, watershed characteristics, reservoir properties, gate and spillway configuration, and water routing relationships. Together, these parameters form the boundary conditions that our CPS-enabled gate control program must respect.

The dam's physical geometry is defined by a height ranging from 149 to 178 ft and a crest length of 2,920 ft, consisting of a composite concrete and earthen embankment construction, a setup common among mid-century USACE dams in the northeastern United States. The contributing watershed spans approximately 1,514-1,600 mi^2, with a mean annual streamflow of 2,962 cfs and a recorded peak of 7,792 cfs, which proves the intensity of storm climate in the region.

The reservoir behind the dam has a total storage capacity of 223,722 acre-ft at normal pool, a surface area of approximately 3,490 acres, and a pool length of 10-11 miles. The normal operating pool elevation ranges from 806 to 809 ft, with roughly 40 ft of fluctuation range. The elevation-storage relationship used in the routing model is summarized below:

Elevation-Storage Curve:

Elevation (ft):	Storage (acre-ft):
769 (lowest possible elevation)	0
779	55000
789	110000
799	170000
809 (middle-high end of pool)	223722

Gate and discharge capacity is characterized by four total control gates (including one sluice gate), an emergency spillway rating of 143,000 cfs, and a maximum safe discharge of 211,000 cfs. Under normal turbine-only operation (gates closed), the dam passes approximately 13,300 cfs, with a maximum turbine capacity of 18,300 cfs. Spillway activation begins near the crest of the dam. The standard use spillways were estimated to have a capacity of 50,000 cfs. At 810 ft the structure releases roughly 40,000 cfs, at 815 ft approximately 100,000 cfs, and at 820 ft the full rated maximum of 211,000 cfs. The spillway is the main physical aspect of the entire system. Small changes in the pool stage near the crest produce large changes in outflow via the gates and spillway.

For the watershed's rainfall-runoff relationship, the most important parameters are the time of concentration of 15-30 hours, a lag time of 12 hours (720 minutes) representing the delay between peak rainfall intensity and peak inflow to the reservoir, and a Soil Conservation Service Curve Number (CN) of 70. This number alone indicates our region has high runoff potential and limited infiltration capacity across the entire watershed, which is largely why we chose this dam as the location for our system.

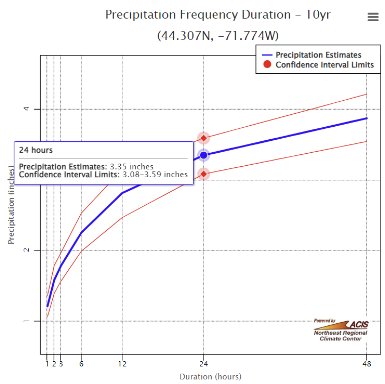
Hydrographs

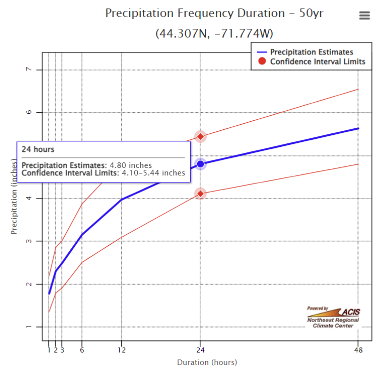
To formulate the basis for modeling the response of the watershed surrounding Moore Dam to weather events of varying magnitude, precipitation data was collected for Littleton, NH, from the NRCC website. This information is vital for being able to model the extent of the design necessary for the gate control and spillway infrastructure. The data gathered pertained to 10-year, 50-year, and 100-year storm events, which refer to the likelihood of events of these severities occurring in a given year. In this case, the storms used in the model would occur, on average, once every 10, 50, and 100 years, respectively.

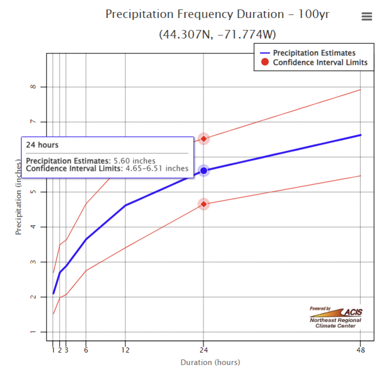
In Figure 1 below, graphical representations of the time series data can be seen, which highlights the precipitation levels over the course of the 24 hours in the design storm. While the curves generally look similar, it is worth noting that the confidence bounds for the 50 and 100-year storms are visibly wider, which is the case because there have naturally been less instances of these larger storms in the historical database.

Figure 1

NRCC time series graphs of precipitation over the course of the 10, 50, and 100 year storms, with confidence bounds and estimates at the 24 hour mark



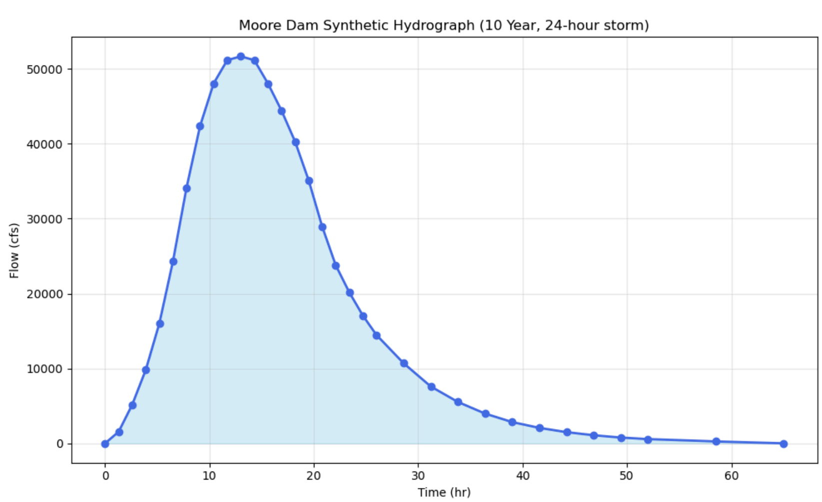




A Python program was then written to convert the precipitation to an inflow rate to the reservoir with units of cubic feet per second. Additionally, incremental time steps of 1.3 hours were used to measure the change in flow rate throughout the entirety of the duration of the retention time. Using these points, a synthetic hydrograph was created for each 24 hour storm, which was derived from an iteration of unit hydrographs for each time step in the process.

Figure 2

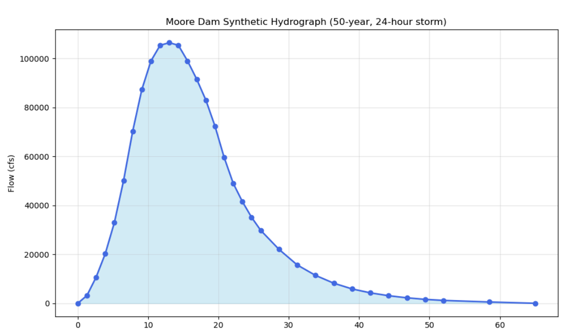
Hydrograph of the 10-year storm event, showing the flow of the stormwater as it accumulates in the watershed and the pattern of outflow via the dam and spillways

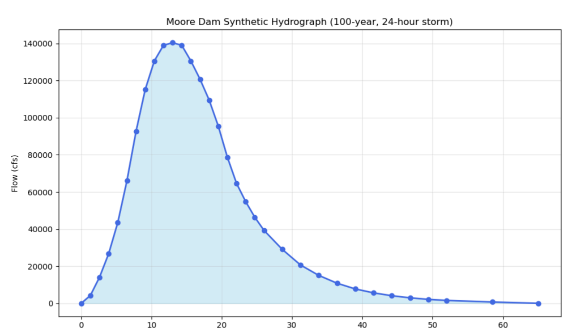


As can be seen in Figure 2, the peak of the flow due to the precipitation event is roughly 12 hours into the storm. This peak value of approximately 51,000 cfs quickly begins to decrease as precipitation decreases towards the end of the 24 hour period. However, in the case of Moore Dam, the 12 hour lag period causes a delay in the release of this storm water, which is highlighted by the tail to the right of the spike in the hydrograph. Also, the fact that the water will take over 60 hours to be fully drained after the 10-year event is important because it influences the need to design the gate system in a manner that accounts for this delay. Figure 3 depicts the two higher storm severities, showing that, most importantly, the maximum flow rate into the watershed will be much higher for the 50 and 100 year events(roughly 106,000 and 140,000 cfs, respectively), meaning that the system must have a significantly higher water capacity than what would be adequate simply for the 10-year event. However, despite having a scaled up flow rate, the overall curve of the hydrograph is very similar, meaning that it can be expected that the drainage patterns would remain consistent over time in each of the storm types.

Figure 3

Synthetic hydrographs for 24 hour, 50 and 100-year storms





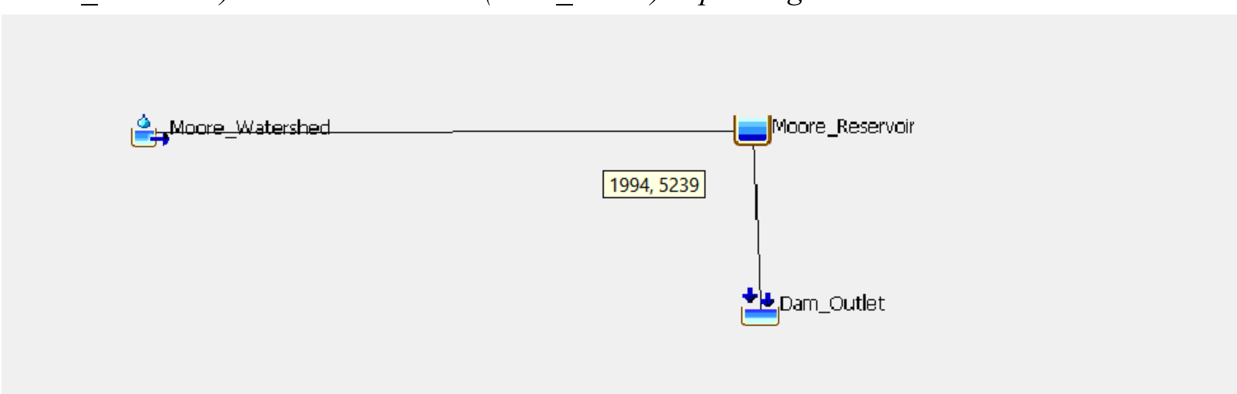
HEC-HMS Baseline Modeling

To operationally work these parameters and simulate the dam under design storm conditions, the watershed-reservoir system was made in HEC-HMS (Hydrologic Engineering

Center - Hydrologic Modeling System), the U.S. Army Corps of Engineers' industry-standard hydrologic modeling platform. The model layout, shown in Figure 4, consists of three connected elements: a subbasin element (Moore_Watershed) representing rainfall-runoff transformation across the 1,514 mi² drainage area, a reservoir element (Moore_Reservoir) applying the elevation-storage and elevation-outflow curves and functions for flood routing, and a downstream sink (Dam_Outlet) recording total outflow from the structure, including the dam's emergency spillway. NRCC precipitation frequency estimates for the project region were used to generate 24-hour design storm inputs for the 10-year, 50-year, and 100-year return period events (see hydrograph section). All simulation runs have data gathered for 2-day periods (48 hours), to understand the longer-term effects of the inflow and outflow before and after the main 24-hour storm events.

Figure 4

Simple Mass-Balance Model for Moore Dam: water flowing into the dam (Moore_Watershed → Moore_Reservoir) exits out the outlet (Dam_Outlet) depending on watershed conditions.



LP optimized dispatch:

This section develops a LP optimization model for hydropower dispatch of a reservoir dam under multiple storm loading scenarios. The model is implemented in Julia using the JuMP mathematical programming framework with Gurobi as the solver, and is evaluated across three storm events, the 10-year, 50-year, and 100-year return period flood derived from HEC-HMS rainfall-runoff simulations for flow rates and rainfall accumulation.

At each timestep, the optimizer allocates inflow across three release pathways: turbine flow, a controlled spillway, and an emergency spillway. The objective function maximizes total energy generation over the storm horizon while penalizing any use of the emergency spillway, incentivizing the model to release water via spillways in advance of the storm to avoid triggering emergency infrastructure. Physical constraints include reservoir mass balance, storage bounds between dead pool and maximum pool elevation, turbine and spillway capacity limits, and ramp rate restrictions on both turbine and controlled spillway flow to reflect real operational limitations.

Results for each storm scenario are reported as a full time step-by-timestep release schedule including turbine flow, controlled spillway discharge, emergency spill volume, reservoir storage, pool elevation. Outflow hydrographs are also generated for each scenario to visually assess the flow rates and differentiate between turbine output spillway output and emergency spillway usage. The framework is designed to be modular, allowing dam parameters and storm inputs to be updated independently, making it suitable as a planning tool for evaluating operational strategies under a range of hydrologic conditions and for dams with differing parameters.

LP Model Formulation:

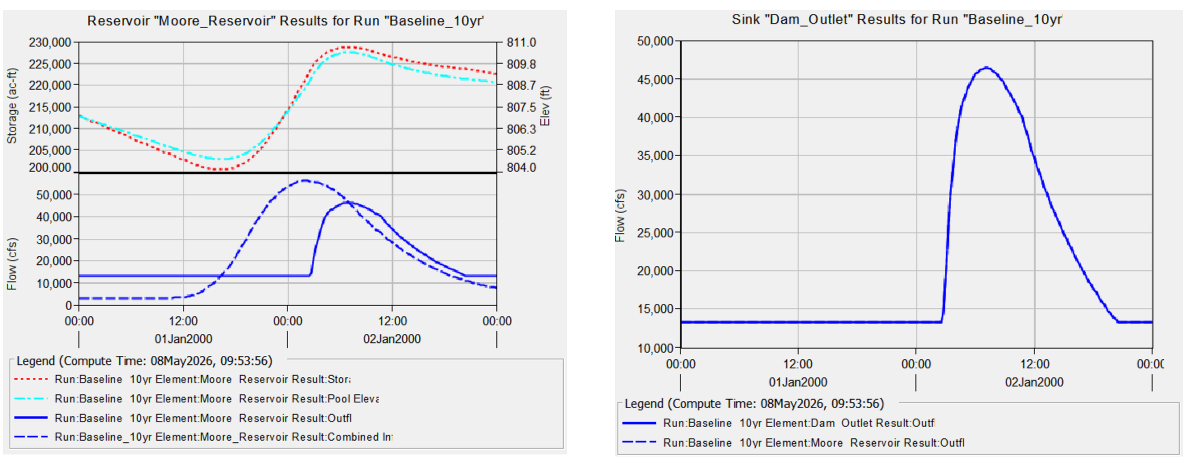
$$
\begin{aligned}
\textbf{max} \quad & \sum_{t=1}^{T} P_t \, \Delta t \;-\; \lambda \sum_{t=1}^{T} Q_t^{\text{emrg}} \\
\text{s.t.} \quad & S_1 = S_0, \\
& Q_1^{\text{spill}} = 0, \\
& S_{t+1} = S_t + \alpha_{\text{conv}} \Delta t \big(Q_t^{\text{in}} - Q_t^{\text{turb}} - Q_t^{\text{spill}} - Q_t^{\text{emrg}}\big), && \forall t, \\
& Q_t^{\text{turb}} + Q_t^{\text{spill}} + Q_t^{\text{emrg}} \le \overline{Q}^{\text{total}}, && \forall t, \\
& Q_t^{\text{spill}} \ge \alpha^{\text{lo}} (S_t - S^{\text{norm}}), && \forall t, \\
& Q_t^{\text{spill}} \ge \alpha^{\text{hi}} (S_t - S^{\text{norm}}), && \forall t, \\
& P_t = \kappa_0 Q_t^{\text{turb}} + \kappa_1 (w_t - S^{\text{ref}} Q_t^{\text{turb}}), && \forall t, \\
& |Q_t^{\text{turb}} - Q_{t-1}^{\text{turb}}| \le \rho^{\text{turb}}, && \forall t \ge 2, \\
& |Q_t^{\text{spill}} - Q_{t-1}^{\text{spill}}| \le \rho^{\text{spill}}, && \forall t \ge 2, \\
& S_{T+1} \ge S^{\text{dead}} + 0.10(\bar{S} - S^{\text{dead}}), \\
& 0 \le Q_t^{\text{turb}} \le \overline{Q}^{\text{turb}}, && \forall t, \\
& 0 \le Q_t^{\text{spill}} \le \overline{Q}^{\text{spill}}, && \forall t, \\
& 0 \le Q_t^{\text{emrg}} \le \overline{Q}^{\text{emrg}}, && \forall t, \\
& S^{\text{dead}} \le S_t \le \bar{S}, && \forall t, \\
& P_t \ge 0, && \forall t.
\end{aligned}
$$

Model Results & Conclusion:

Results from the three baseline HEC-HMS runs, with no gate operations and no forecast-based pre-release, show a concerning pattern of increasing stress on the dam system as the storm return period increases. For the 10-year event, dam outlet flow holds steady at the baseline turbine release of approximately 13,300 cfs throughout the first day as inflow slowly builds in the watershed (Figure 5). Around midnight, inflow to the reservoir rises rapidly to trigger the spillway, and peak outflow at the spillway reaches approximately 46,500 cfs in the early hours of the next day before receding. The reservoir pool climbs to roughly 810.5 ft during this event, largely above the pool, and storage peaks near 227,000 acre-ft.

Figure 5

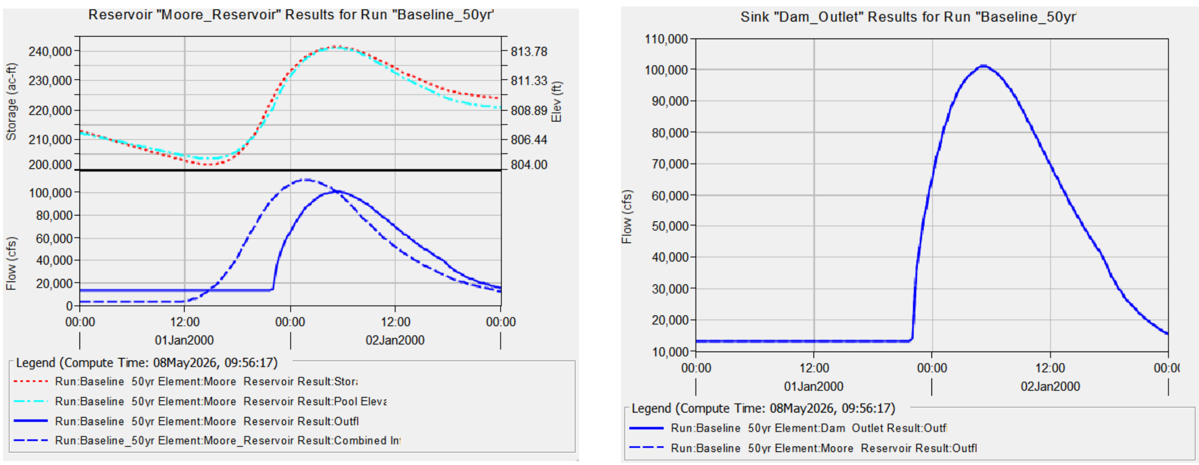
10-year simulation results for Moore Dam. The spillway is activated around midnight as inflow from the watershed overtops the normal turbine release capacity, producing a peak outlet discharge of approximately 46,500 cfs before the pool recedes.



For the 50-year event, the pattern is similar but more severe (Figure 6). The gate holds at turbine-only discharge through the first day, then spillway activation around midnight drives peak outflow at the spillway to approximately 101,000 cfs, roughly seven times the baseline turbine release (13,300cfs). The reservoir pool elevation reaches approximately 813.8 ft, and storage climbs to nearly 240,000 acre-ft, way above the limits of the reservoir pool.

Figure 6

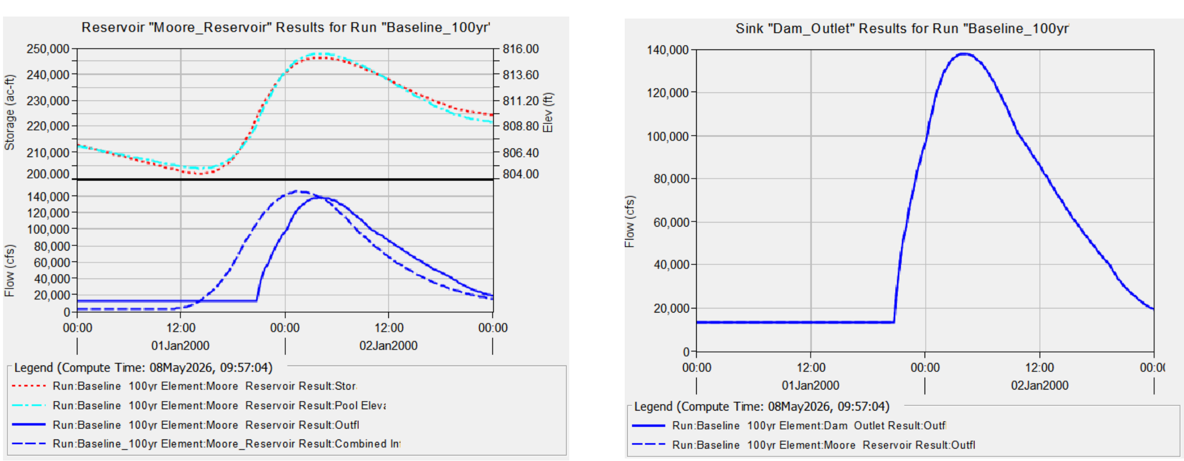
50-year simulation results for Moore Dam. The spillway is activated near midnight on the first day as combined watershed inflow reaches approximately 105,000 cfs, driving peak dam outlet discharge to roughly 101,000 cfs and pushing pool elevation to approximately 813.8 ft.



The 100-year event represents the most severe baseline scenario (Figure 7). Combined inflow to the reservoir peaks at approximately 140,000 cfs, and the reservoir pool climbs to an elevation near 816 ft, roughly 7 ft above normal pool and approaching the upper portion of the spillway operating range. Storage peaks at approximately 248,000 acre-ft, and peak dam outlet discharge reaches approximately 137,000 cfs in the early hours of the second day.

Figure 7

100-year simulation results for Moore Dam. The spillway is activated around midnight as inflow peaks near 140,000 cfs, with pool elevation at approximately 816 ft and peak outlet discharge roughly at 137,000 cfs. This is by far the most severe result across all three baseline scenarios.



These results create the reference condition against which the CPS-logic forecast scenario will be compared. The critical finding is that all storms overwhelm the static gates. The reservoir fills faster than the turbines of the dam can handle, which will force the emergency spillway to be used and produce high peak outflows that could affect areas greater than just the dam. This motivates the forecast LP algorithm described in the following section. The next three figures describe scenarios where the LP forecast/dispatch optimization code activates gates and other spillways to limit the usage of the emergency spillway and turbines from the different storm intensities.

Figure 8

Outflow through the dam over 48 hours during a 10 year storm compared to rainfall (inflow).

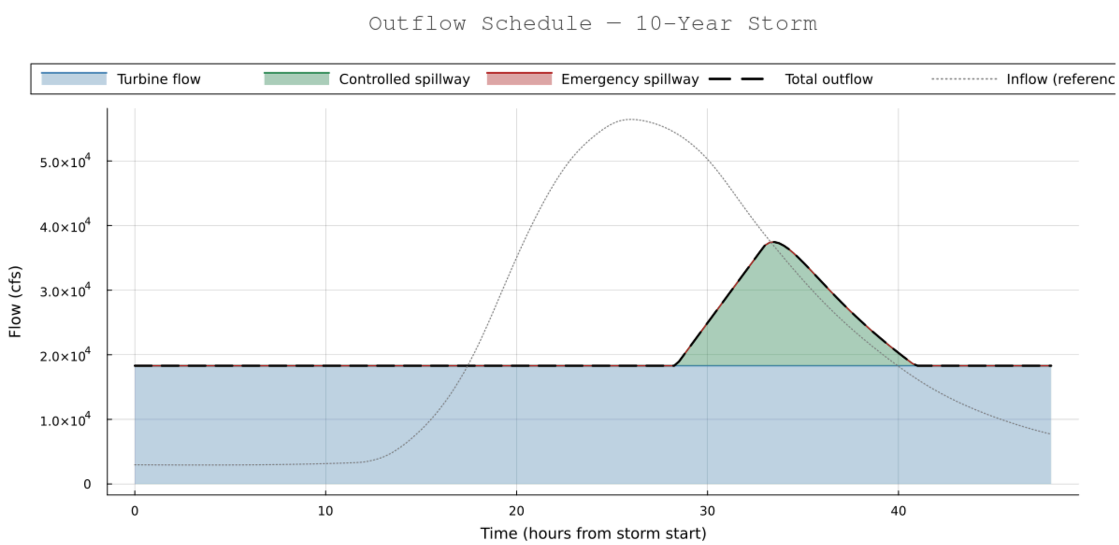


Figure 9

Outflow through the dam over 48 hours during a 50 year storm compared to rainfall (inflow).

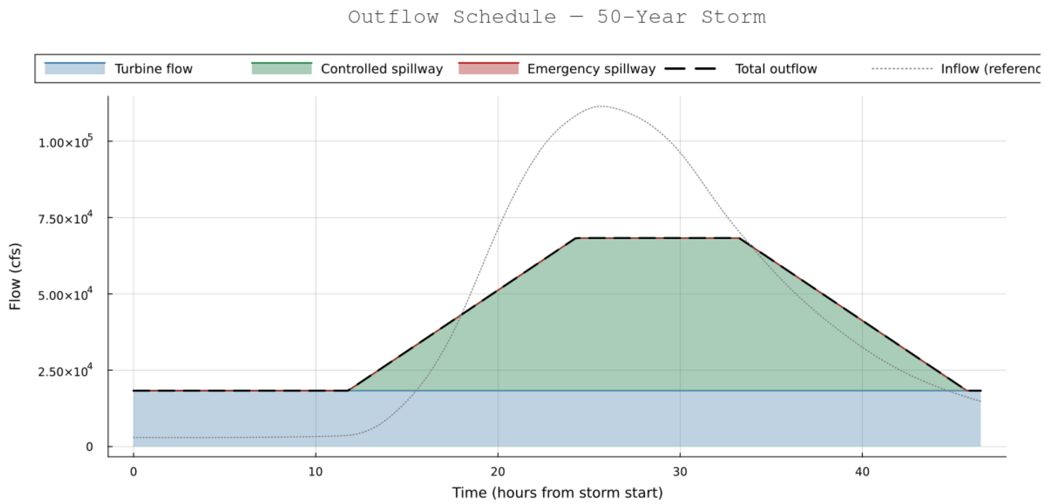
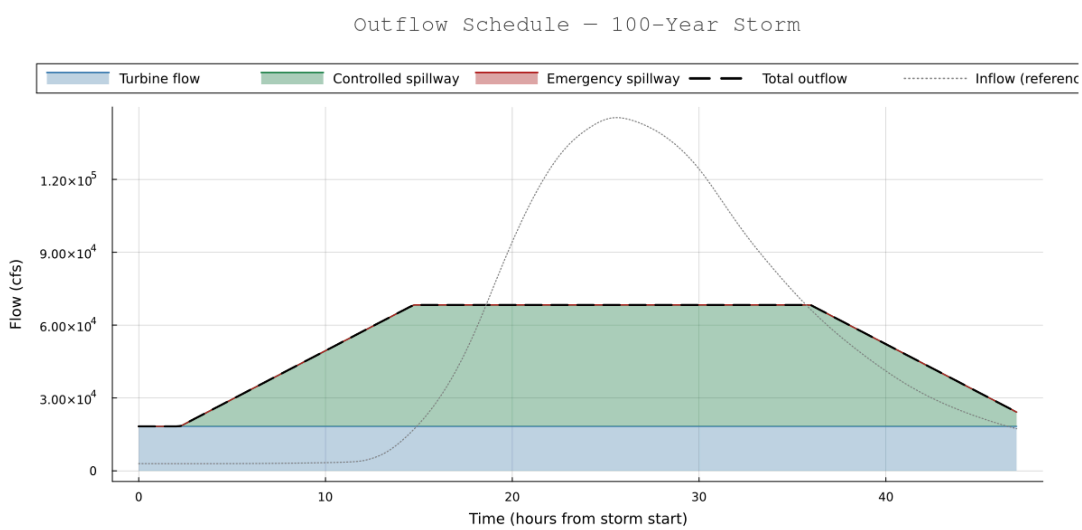


Figure 10

Outflow through the dam over 48 hours during a 100 year storm compared to rainfall (inflow).



The results of this study demonstrate that a tiered LP dispatch model can simultaneously maximize hydropower generation and prevent emergency spillway activation across all three design storm events. By introducing a controlled spillway as a dedicated intermediate release pathway, the optimizer successfully routed the 10-year, 50-year, and 100-year storms through the reservoir without triggering any emergency spill infrastructure. Turbine flow was held at or near the 18,300 cfs maximum throughout each event, while maximum pool elevations of 809.27, 809.62, and 809.69 ft remained well below the 811.0 ft emergency threshold in all cases. The 5,000 cfs per timestep ramp constraint on the controlled spillway produced smooth, gradually rising hydrographs, most evident in the 100-year scenario where the spillway begins opening at hour 2 ¼ and sustains its 50,000 cfs capacity for over 20 hours, reducing peak discharge relative to an unoptimized model.

Several limitations should be acknowledged. The basis of this optimization relies on perfect forecast information about 12 hours in advance of the beginning of the storm. The single-event HEC-HMS inflows do not capture multi-storm sequences or any other compounding effects. Future work incorporating forecast uncertainty through a Monte Carlo simulation could help optimize the system even better. There are also minor things, like a small penalty for running the turbines above the preferred 13,300 cfs, which could produce wear on the turbine over time. Nonetheless, the model confirms that properly controlled gate release infrastructure, paired with a forecast optimization-based dispatch strategy, can eliminate emergency spillway reliance even under the 100-year design event while recovering hydropower value from flood inflows. By spreading out this flow, decreasing the peaks substantially, we should be able to decrease long term erosion, and other stress effects on the dam.

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